A Web-Based Workflow for Variable-Star Light Curves: Exploring and Processing TESS Data

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Tatra Astro Summit 2026

SLOVAK RESEARCH AND DEVELOPMENT AGENCY
This research has been supported by the grant of the Slovak Research and Development Agency APVV-20-0148

Motivation

"What do we have for this object, and what does it look like?"

User-friendly web portal for variable-star research

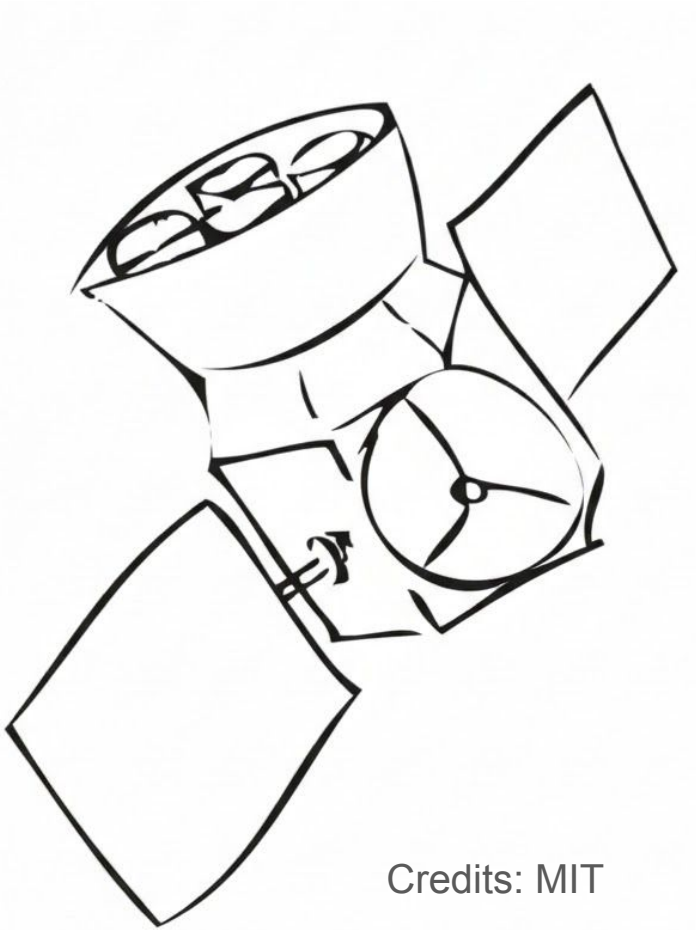
data discovery → retrieval → processing → analysis



TESS part: `Lightkurve` open-source Python package
Lightkurve Collaboration (2018)

Web portal: Dash (Plotly)

TESS refresher



Credits: MIT

TESS refresher: basic facts

***T**ransiting **E**xoplanet **S**urvey **S**atellite*

Launch: April 2018
Sector 1: 25 July - 22 August 2018

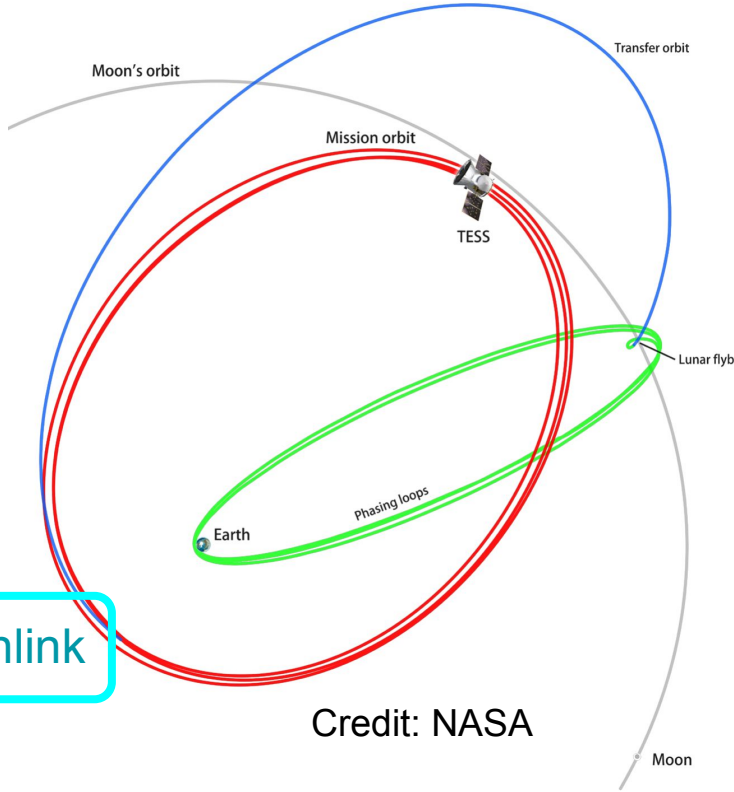
Orbit: geocentric, highly elliptical
Period: 13.7 days, 2:1 resonance with the Moon
1 Sector = two orbits = 1 sidereal month*

Cameras: f/1.4 D≈10.5 cm (targets bright stars T≲13)

Bandpass: Red, 600-1000 nm

Pixel scale: 21″/px

*true for most sectors; see next slide

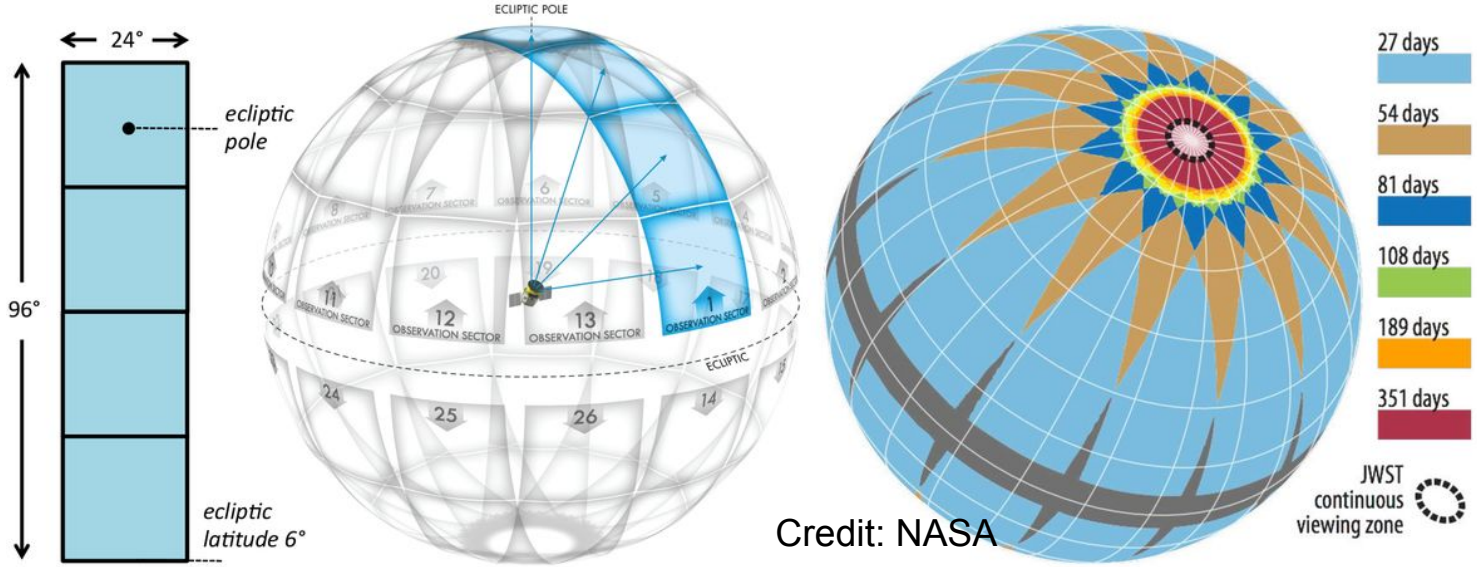


downlink

Credit: NASA

TESS refresher: Sectors

Sector = Sky field × Time window



Credit: NASA

Current Status: Sector 108 (September 7 - October 5, 2026)
The TESS data has **no proprietary time**

TESS refresher: Products

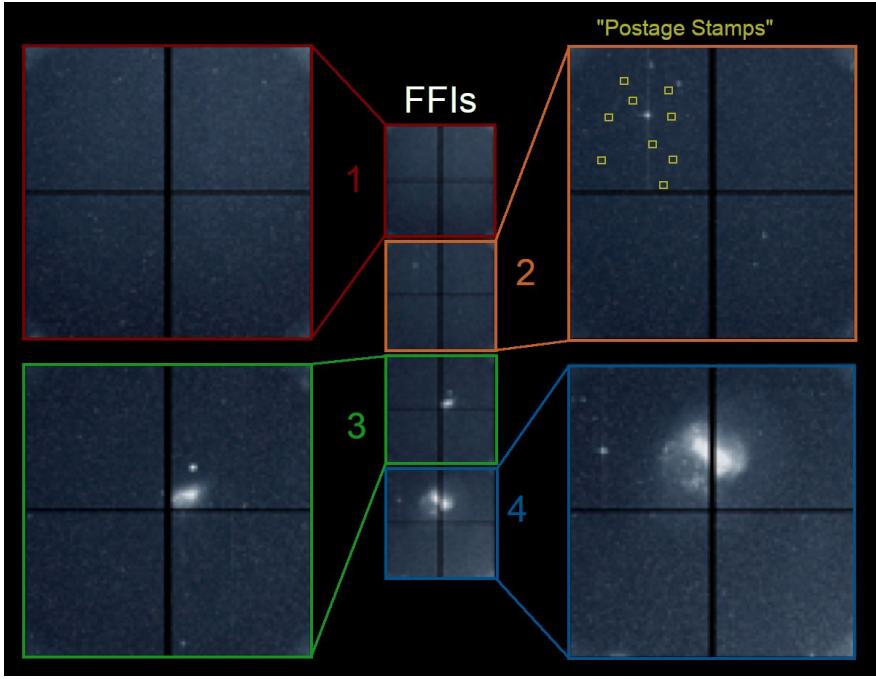
Pipeline lightcurves: produced from TPF or FFI by SPOC, TESS-SPOC, QLP, and other pipelines

FFI *(Full Frame Image):*
cadence: 1800 / 600 / 200 sec

TPF *(Target Pixel Files)*: cutouts around selected targets; *background-subtracted, secondary CR*
on-board cadence: 120 sec or 20 sec (since 2020)

All of these come from raw 2-second images (that never reach the ground)

cosmic-ray mitigation (CRM) is applied onboard (except 20 sec TPF cadence)

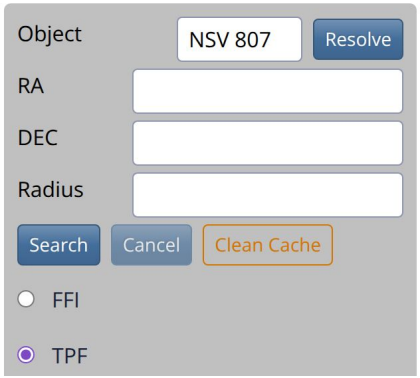




https://astronomy.science.upjs.sk/igebc/tess
https://astronomy.science.upjs.sk/igebc/tess_lc

Workflow Demonstration

Workflow: From cutouts – search



TPF

#	Mission	Year	Author	Exptime
1	TESS Sector 04	2018	TESS-SPOC	1800
0	TESS Sector 30	2020	SPOC	120
2	TESS Sector 30	2020	TESS-SPOC	600

FFI

TESS Sector 04	2018	TESScut	1425.599419	NSV 807
TESS Sector 30	2020	TESScut	475.199788	NSV 807
TESS Sector 97	2025	TESScut	158.399925	NSV 807

CRM algorithm discards 20% – 2 values from every ten 2 sec raw exposures

Cadence vs exposure mystery

600s FFI:
Integration_Time = 1.98s
Read_time = 0.02 s,
N_Readout = 300 * 0.8 = 240 (80% discarded by CRM)

Exposure = N_Readout x Integration_Time

Exposure = 240 * 1.98s = 475.2s
Cadence = 300 * 2.00s = 600.0s

120s TPF:
Exposure = 60 * 0.8 * 1.98s = 95.4s
Cadence = 60 * 2.00s = 120.0s

Workflow: From cutouts – cutout

TESS Cutout Tool

Search Sector | Plot

Cutout Tools

▼ Plot options

Scale 2

Sum

Frame

0 — 18354

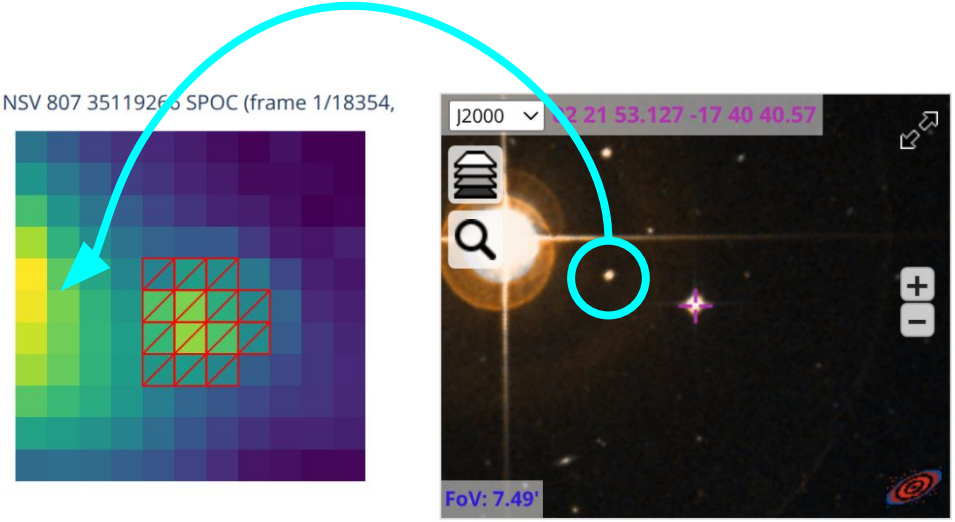
rePlot pixel

▼ Mask

- Auto
- Handmade
- pipe

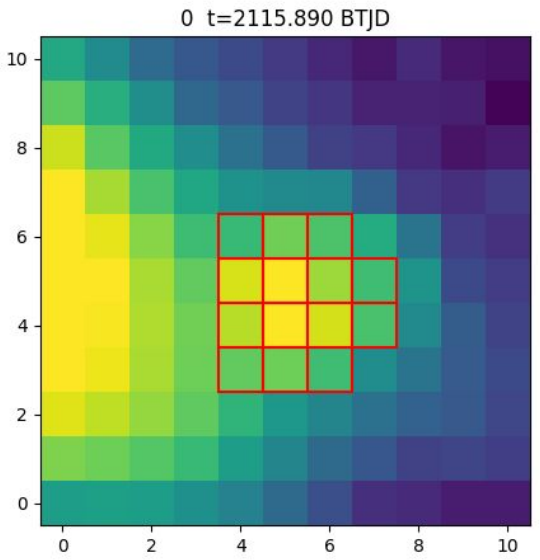
Select mask and build the lightcurve:

- Click on a star in the **Aladin** applet to mark it on the pixel image
- **Handmade Mask:** Click on a pixel to set/unset mask
- **Auto-mask:** Click on a pixel to create a threshold mask around it
- **Pipeline mask:** Use the mask provided by the team



Workflow: From cutouts – result

Extracted lightcurve of the eclipsing binary NSV 807



Curve Tools

Sub bkg

Flatten

MJD

Date

rePlot curve

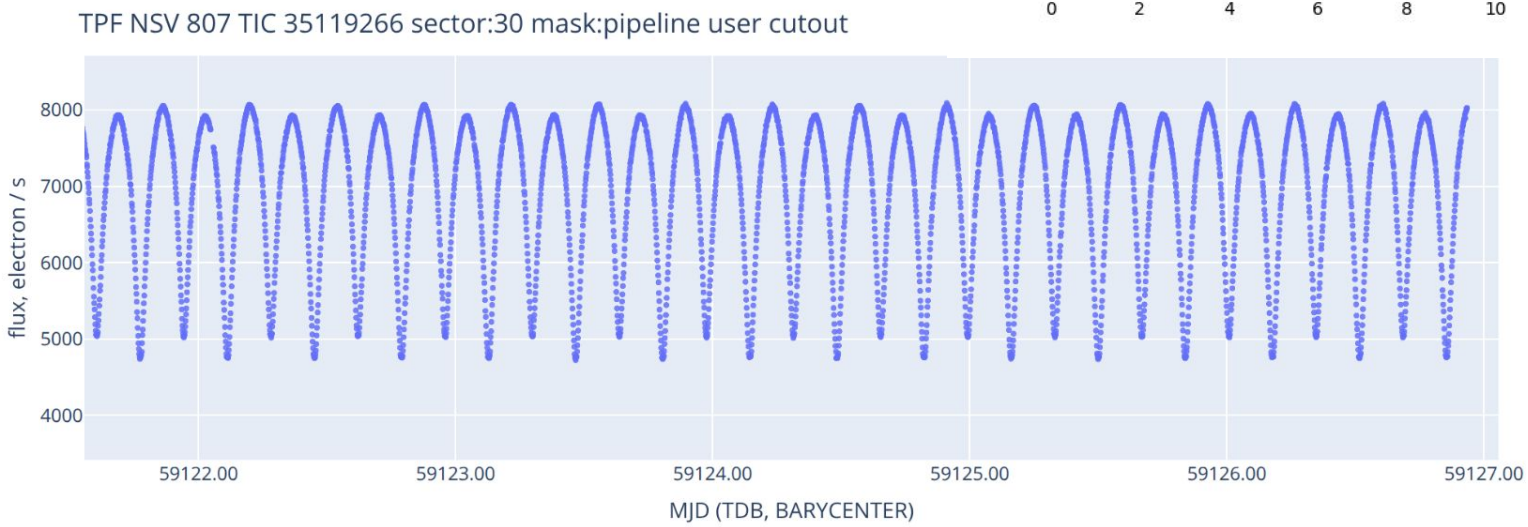
▶ Plot Options

Trim selected

VOT

Download

First Light Curve



Workflow: Pipelines



TESS Lightcurve Tool

Search | Plot

Object: NSV 807 [Resolve]
RA:
DEC:
Radius: 5

[Search] [Cancel] [Clean Cache]
[Upload] Append

NSV 807 (TIC 35119266)

[Retrieve curves]

#	Mission	Year	Author	Exptime
3	TESS Sector 04	2018	QLP	1800
6	TESS Sector 04	2018	GSFC-ELEANOR-LITE	1800
7	TESS Sector 04	2018	TARS	1800
9	TESS Sector 04	2018	TASOC	1800
10	TESS Sector 04	2018	TASOC	1800
11	TESS Sector 04	2018	TGLC	1800
0	TESS Sector 30	2020	SPOC	120
2	TESS Sector 30	2020	TESS-SPOC	600
4	TESS Sector 30	2020	QLP	600
8	TESS Sector 30	2020	TARS	600
5	TESS Sector 97	2025	QLP	200

Workflow: Pipelines – Who is Who

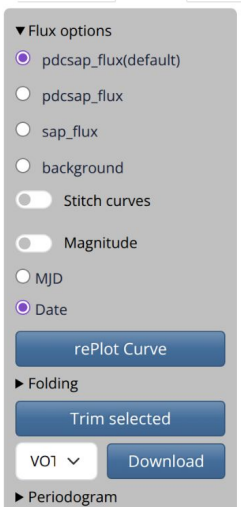


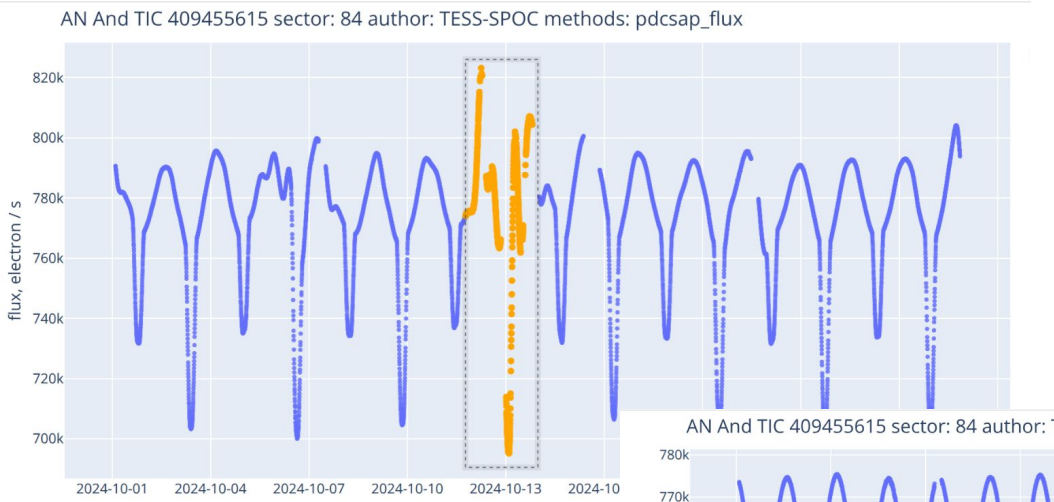
SPOC – ***S**cience **P**rocessing **O**perations **C**enter* (NASA Ames):
source: 120s / 20s on-board TPFs
for the preselected target list (~20 000 per sector)

TESS-SPOC: the same pipeline
source: TPFs built on the ground from FFI cutouts (30 min / 10 min/ 200s)
for extended target list (~160 000 per sector)
TESS Science Data Products Description

QLP – ***Q**uick-**L**ook **P**ipeline* (MIT)
source: FFIs
magnitude limit Tmag <= 13.5
QLP changed the extraction algorithm starting from Sector 94
see Quick-Look Pipeline

Workflow: Pipelines – SAP / PDCSAP

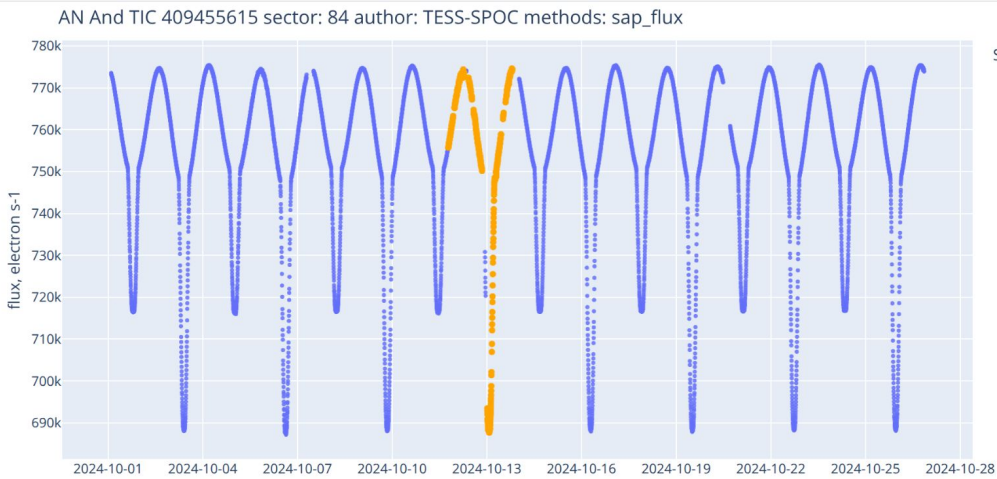




PDCSAP

***P**research*
***D**ata*
***C**onditioning*
(default)

SAP



Workflow: Pipelines – SAP / PDCSAP



SAP – ***S**imple **A**perture **P**hotometry*

- sum inside an aperture mask

PDCSAP – ***P**research **D**ata **C**onditioning SAP*

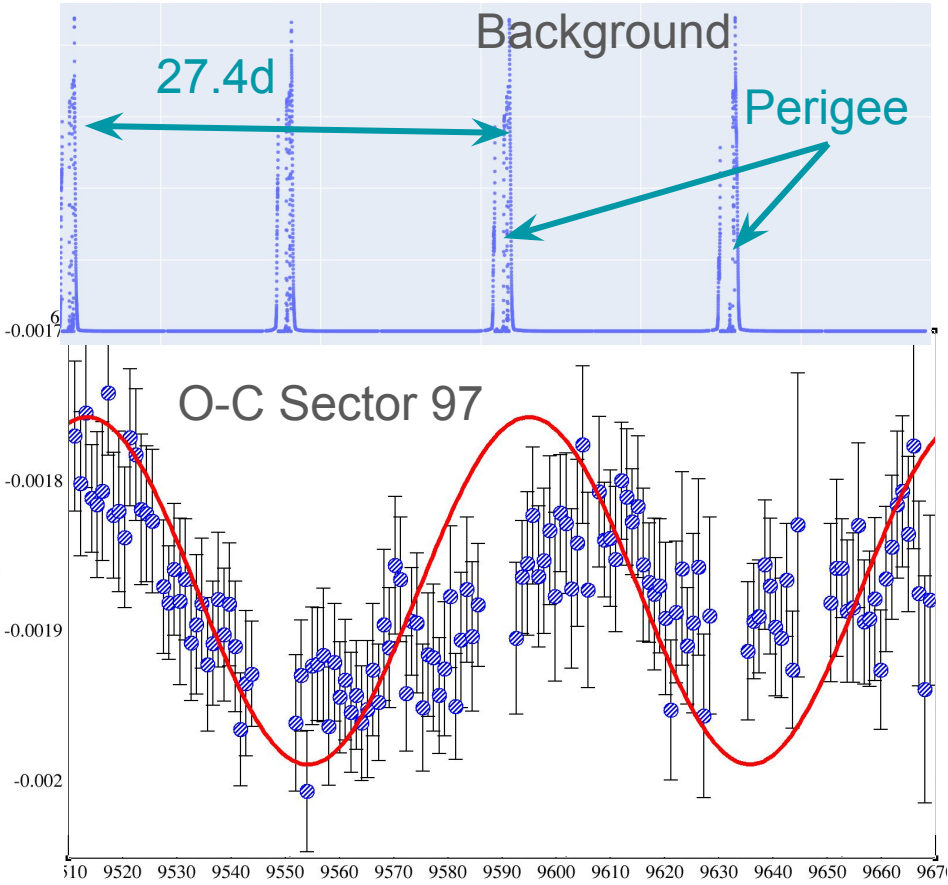
removes systematic trends using Cotrending Basis Vectors (CBVs)

a per-detector-channel, per-observing-period set of the most common trend shapes seen across many quiet stars

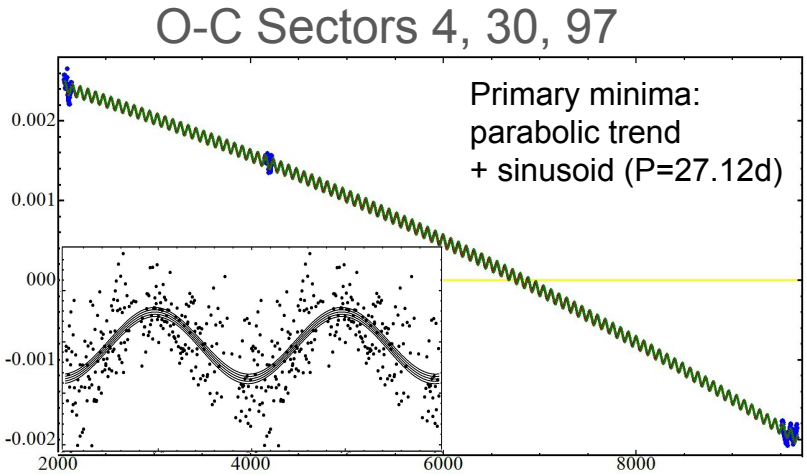
NASA Kepler Guest Observer Program

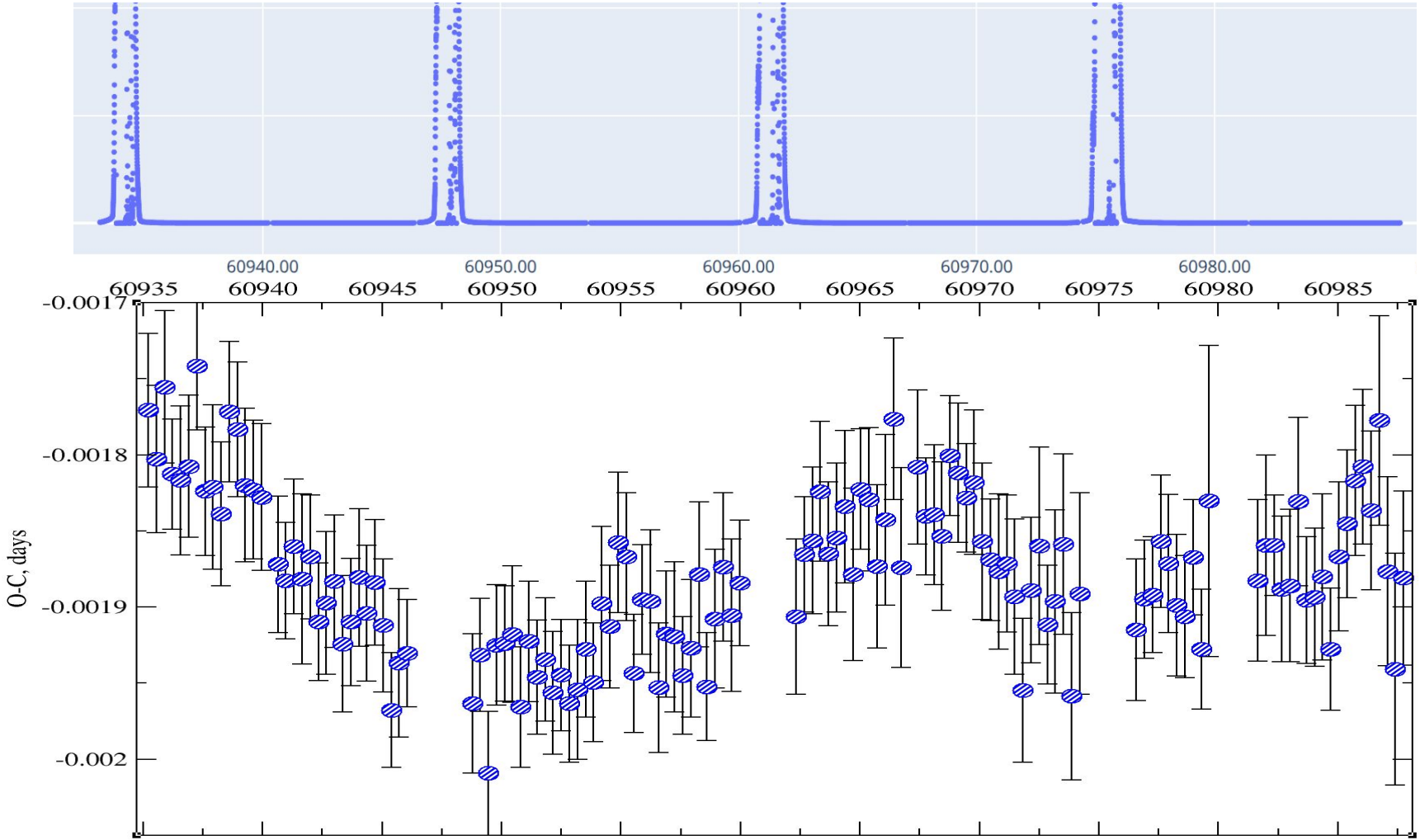
KSPSAP_FLUX – QLPs detrended flux

Workflow: Pipelines – Background

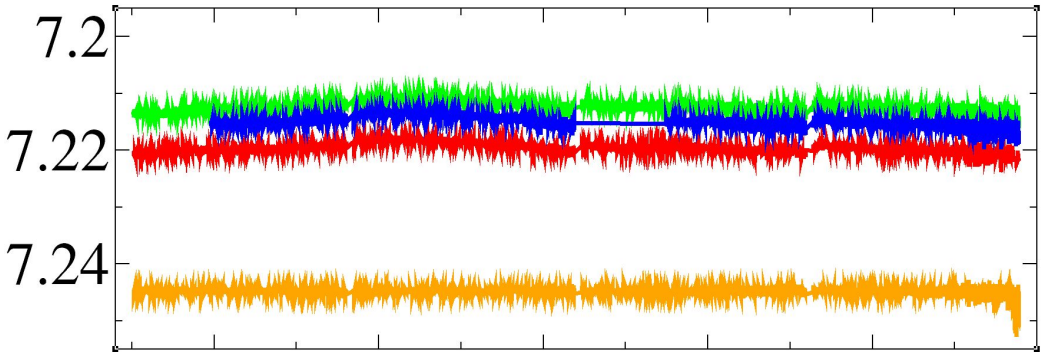


NSV 807
eclipsing binary




60940.00
60950.00
60960.00
60970.00
60980.00
60935
60940
60945
60950
60955
60960
60965
60970
60975
60980
60985
-0.0017
-0.0018
-0.0019
-0.002
O-C, days

TESS Photometric Calibration?



HD 37962

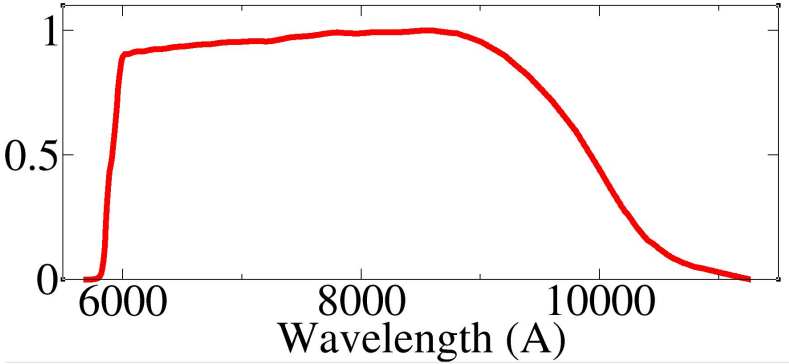
- from FFI
- QLP
- SPOC
- from TPF

flux[e-/s] → Tmag	Tmag → Flux[Jy]
Tmag = 20.44[±0.05] - 2.5lg(flux)	Flux[Jy] = 2632 * 10^(-0.4*Tmag)

Real calibration uncertainties > 20% (>100% for faint) (TESS Transients)

Nevertheless, ***we provide exported lightcurves with VO-compliant photometric calibration***

TESS/TESS.Red



Filter Profile Service “Carlos Rodrigo”

The portal is online.
Your feedback is very welcome

https://astronomy.science.upjs.sk/igebc/tess



Thank you for your attention!

References

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2. Lightkurve Collaboration et al., 2013
   Lightkurve: Kepler and TESS time series analysis in Python,
   Astrophysics Source Code Library, record ascl:1812.013

3. NASA Kepler Guest Observer Program, 2012
   PyKE Primer – 4. Light Curve Files
   https://nexsci.caltech.edu/workshop/2012/keplergo/PyKEprimerLCs.shtml

4. Plotly Technologies Inc. *Dash documentation*. https://dash.plotly.com

5. Rodrigo, C., Solano, E., 2020
   The SVO Filter Profile Service
   Contributions to the XIV.0 Scientific Meeting (virtual) of the Spanish Astronomical Society,
   held 13-15 July 2020

References (continue)

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   https://archive.stsci.edu/missions-and-data/tess

7. *TESS Instrument Handbook*
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8. *TESS Science Data Products Description Document*
   https://archive.stsci.edu/missions/tess/doc/EXP-TESS-ARC-ICD-TM-0014.pdf

9. TESS Science Office (MIT)
   Quick-Look Pipeline (QLP).
   tess.mit.edu/qlp

10. MIT / Fausnaugh et al.
    TESS Transients. README
    https://tess.mit.edu/public/tesstransients/pages/readme.html